

Conductivity Type Water Level Controller CPLS-1700L

PRODUCT DESCRIPTION:



The CPLS-1700L Conductivity Level Switch is a compact and reliable instrument for point-level detection of conductive liquids. It detects liquid level through electrical conductivity between the sensing probe and reference electrode, generating a switching output when the liquid reaches the preset level.

Available in Integral and Remote Type configurations, the CPLS-1700L offers rigid/flexible probe options and can be customised for different process conditions. It is suitable for tank level monitoring, pump control, high/low-level alarms and overflow protection.

No moving parts, simple construction and low maintenance make it suitable for a wide range of industrial liquid-level applications.

OPERATING PRINCIPLE:

Conductivity Level Switches operate on the principle of electrical conductivity to provide reliable point-level detection of conductive liquids. A low AC voltage is applied between the sensing probe(s) and the reference electrode or conductive tank wall. When the conductive liquid reaches the sensing probe, it completes the electrical circuit between the electrodes. The resulting change in conductivity is detected by the electronic circuitry, which generates a switching signal corresponding to the detected level. The instrument can be configured for single-point or multi-point level detection, allowing functions such as High-Level, Low-Level, Pump ON/OFF control, overflow protection, dry-run protection, and alarm generation. The use of low AC sensing voltage minimizes electrode polarization and supports stable, reliable operation. Since there are no moving mechanical parts, the instrument provides long service life with minimal maintenance. Conductivity Level Switches are suitable for conductive liquids with conductivity typically $\geq 20 \mu\text{S}/\text{cm}$, including water, treated water, wastewater, chemicals, acids, alkaline solutions, and other conductive process liquids. The switches are available with rigid or flexible probes in various lengths and materials to suit different tank depths, mounting arrangements, and process conditions. Standard versions are suitable for temperatures up to 120°C and pressures up to 10 bars. Special high-temperature and high-pressure versions are available for temperatures up to 250°C and pressures up to 20 bars.

SALIENT FEATURES

- Reliable point-level detection for conductive liquids $\geq 20 \mu\text{S}/\text{cm}$
- No moving parts – maintenance-free operation
- Integral & Remote Type available
- Rigid & Flexible Probe options
- Single / multi-point level detection
- High / Low Level & Pump Control applications
- SS304 / SS316 probe options
- Temperature up to 120°C ; special up to 250°C
- Pressure up to 10 bar; special up to 20 bar



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- Weatherproof / Flameproof enclosure options
- LED indication for easy status monitoring
- Customised design as per application requirements

PROBE CONSTRUCTION:

The CPLS-1700L is provided with robust stainless-steel sensing probes designed for reliable conductivity-based level detection. SS304 / SS316 probe options are available in rigid and flexible configurations with customised probe lengths.

- Rigid / Flexible Probe construction
- SS304 / SS316 material options
- Various probe lengths available
- Suitable for single / multi-point sensing
- PTFE insulation available as required
- Suitable for Integral & Remote Type installation
- Customised probe construction available for specific process conditions

TECHNICAL SPECIFICATIONS – CPLS-1700L

Parameter	Specification
Model	CPLS-1700L
Make	Levtron Instruments Pvt. Ltd
Type	Remote Electronics
Power Supply	230 VAC / 24 VDC – Optional
Output	2 × SPDT, 5A Potential-Free Relay
Sensitivity	Water & Conductive Liquids
Minimum Conductivity	$\geq 20 \mu\text{S/cm}$
Controller Enclosure	ABS, DIN Rail Mounting, Indoor Installation
Level 1	Single Point / Pump Control Logic
Level 2	Second Point / Pump Control Logic
Reference / Ground	Length as per application
Indication	LED – Green / Yellow / Red
Process Connection – Single Point	Individual Probe, ½" BSP / NPT onwards
Process Connection – 2 Point	1½" Screwed / Flanged
Alternative Connection	2" HDPE / PTFE, 10 mm Thickness – Optional
Probe Type	Rod / Rope
Probe Length	25 mm onwards, up to 20 m – Specify
Level 1 Probe	Rod / Rope – Length as required
Level 2 Probe	Rod / Rope – Length as required
Reference / Ground Probe	Length as required

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Parameter	Specification
Counterweight	SS316 × 3 Nos. – For Rope Type
Operating Temperature	0 to 120°C
High Temperature Option	Up to 250°C
Operating Pressure	Standard: Up to 10 bar
High Pressure Option	Up to 20 bar
Service	STP, Water & Conductive Liquids
Field Enclosure	Cast Aluminium, Weatherproof
Cable Gland	PG13.5 × 1 No. PVC Gland

Note: Probe length, process connection, probe material and mounting arrangement can be selected according to the application and customer requirement.

Manufactured By:

Levtron Instruments Pvt Ltd

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